
Analyzing Skill Demand and Salary Trends in the Artificial Intelligence Job Market

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Abstract

Artificial Intelligence has become one of the fastest-growing areas in technology, creating strong demand for workers with machine learning, programming, data, and cloud computing skills. However, students and early-career professionals often face difficulty understanding which AI skills are most valuable and what job characteristics affect salary.

In this project, I analyze a dataset of 15,000 AI job postings from the *Global AI Job Market and Salary Trends 2025* Kaggle dataset. The dataset includes job title, salary, experience level, company size, industry, remote work ratio, education requirement, required skills, years of experience, and benefits score. I build a complete data science pipeline that includes data cleaning, exploratory data analysis, feature engineering, salary prediction, feature importance analysis, and K-Means clustering.

The results show that experience level is the strongest factor affecting salary. Entry-level roles have an average salary of about \$63,133, while executive-level roles average about \$187,724. The most demanded skills include Python, SQL, TensorFlow, Kubernetes, Scala, PyTorch, Linux, Git, Java, and GCP. For salary prediction, Random Forest performs better than Linear Regression, achieving an RMSE of 21,408.86 and an R^2 score of 0.8743. K-Means clustering identifies three clear job groups based on salary, years of experience, remote work ratio, and benefits score.

Overall, this project shows that AI salary depends on multiple interacting factors, especially experience, job title, company size, and skills. The analysis also shows that machine learning can provide useful insights into real job market trends.

Code: https://github.com/YOUR_GITHUB_LINK

Video: [YOUR_VIDEO_LINK](#)

1 Introduction

Artificial Intelligence is now used in many industries, including technology, healthcare, finance, retail, education, transportation, and manufacturing. Companies use AI systems for automation, prediction, recommendation, fraud detection, natural language processing, computer vision, and business decision-making. Because of this growth, demand for AI-related jobs has increased quickly.

At the same time, the AI job market can be difficult to understand. Job postings often list many different skills, such as Python, SQL, TensorFlow, PyTorch, cloud platforms, Linux, Kubernetes, and Git. Salaries also vary widely based on experience level, company size, industry, remote work options, and job title. For students preparing for AI careers, it is useful to understand which skills appear most often and which job characteristics are most related to salary.

The main research question for this project is:

What skills are most demanded in AI jobs, and how do job characteristics such as experience level, company size, industry, and remote work influence salary?

To answer this question, this project analyzes 15,000 AI job postings using a full data science workflow. The project focuses on five smaller questions:

- What are the most common AI job titles?
- What skills appear most frequently in AI job postings?
- How does salary change by experience level?
- Can machine learning models predict AI salary from job features?
- Can AI jobs be grouped into meaningful clusters?

This project is useful because it connects data science methods with a real career-related problem. It also shows how exploratory analysis, regression modeling, and clustering can be combined to better understand labor market trends.

2 Related Work

Job market analysis is a common application of data science. Many studies use job posting datasets to understand salary trends, skill demand, and employment patterns. These analyses are helpful because online job postings contain structured and semi-structured information about employer expectations.

Salary prediction is usually treated as a supervised regression problem. In this type of task, the target variable is salary, and the input features may include job title, location, experience level, company size, required skills, and education level. Linear Regression is often used as a baseline because it is simple and easy to interpret. However, salary is not always explained by a simple linear relationship. For example, the salary difference between entry-level and mid-level jobs may not be the same as the difference between senior and executive jobs.

Tree-based models such as Random Forest are useful for this type of problem because they can model nonlinear relationships and interactions between variables. Random Forest builds many decision trees and combines their predictions, which usually improves performance compared to a single tree or simple linear model.

Skill demand analysis is another important part of job market research. By counting skill frequencies in job postings, it is possible to identify which technical tools are most valued by employers. For AI jobs, this often includes programming languages, machine learning frameworks, database tools, cloud platforms, and deployment tools.

Clustering is also useful for labor market analysis because it can group similar jobs without using a target label. K-Means clustering can help identify job segments based on salary, experience, remote work, and benefits. This project combines these methods into one workflow instead of focusing only on salary prediction or only on visualization.

3 Method

3.1 Dataset

The dataset used in this project is the *Global AI Job Market and Salary Trends 2025* dataset from Kaggle. It contains 15,000 AI-related job postings. Each row represents one job posting, and each column describes a job attribute.

Important columns include job title, salary in USD, experience level, years of experience, company size, company location, industry, remote work ratio, required skills, education requirement, benefits score, and job description length.

The main prediction target is `salary_usd`. The main features include experience level, years of experience, company size, industry, remote ratio, benefits score, job title, and skill-related information.

3.2 Data Cleaning

The first step was data cleaning. I loaded the raw dataset using Python and Pandas, checked the number of rows and columns, checked for missing values, and checked for duplicate rows. The final cleaned dataset contained 15,000 rows and 20 columns.

The cleaning process included standardizing column names, checking missing values, checking duplicate records, converting salary into a numeric USD variable, preparing categorical variables for modeling, splitting and cleaning the skills column, and saving the cleaned dataset for EDA and modeling.

Cleaning the skills column was important because each job posting could contain multiple skills. These skills had to be separated so that individual skill frequencies could be counted correctly.

3.3 Exploratory Data Analysis

Exploratory data analysis was used to understand the dataset before modeling. I created visualizations for salary distribution, top job titles, top skills, salary by experience level, salary by company size, salary by remote work ratio, salary by industry, and numerical feature correlations.

The salary distribution showed that most AI jobs fall in a middle salary range, while fewer jobs have extremely high salaries. The dataset had a minimum salary of \$32,519, a median salary of \$99,705, and a maximum salary of \$399,095. The mean salary was approximately \$115,349.

The most common job titles included Machine Learning Researcher, AI Software Engineer, Autonomous Systems Engineer, Machine Learning Engineer, AI Architect, Head of AI, NLP Engineer, Robotics Engineer, Data Analyst, and AI Research Scientist.

Skill frequency analysis showed that the most common skills were Python, SQL, TensorFlow, Kubernetes, Scala, PyTorch, Linux, Git, Java, and GCP.

3.4 Feature Engineering

Feature engineering was used to prepare the dataset for machine learning. Categorical variables such as experience level, company size, job title, industry, and employment type were encoded into numerical form.

Experience level was especially important because it has a natural order. Entry-level, mid-level, senior-level, and executive-level roles were mapped into ordered values so that the models could learn the relationship between seniority and salary.

Numerical variables such as years of experience, remote work ratio, benefits score, and job description length were also included. These features were combined into a final feature matrix for regression modeling.

3.5 Salary Prediction Models

The supervised learning task was to predict salary. I tested two regression models:

- **Linear Regression:** used as a baseline model.
- **Random Forest Regressor:** used as a nonlinear model.

Linear Regression was included because it is simple and interpretable. However, it assumes a mostly linear relationship between features and salary. Random Forest was included because salary may depend on complex interactions between features. For example, the effect of experience level may depend on job title, company size, or industry.

The dataset was split into training and testing sets. The models were trained on the training data and evaluated on the test data.

3.6 Clustering

K-Means clustering was used to group similar AI jobs. The clustering features included salary, years of experience, remote work ratio, and benefits score. These features were selected because they describe important differences between jobs.

K-Means was applied with three clusters. After clustering, each group was analyzed by comparing average salary, years of experience, remote ratio, and benefits score. This helped interpret the job market segments.

4 Evaluation

4.1 Evaluation Metrics

The regression models were evaluated using Mean Squared Error, Root Mean Squared Error, and R^2 score.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

$$RMSE = \sqrt{MSE} \quad (2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

RMSE is useful because it shows the average prediction error in salary dollars. R^2 shows how much salary variation is explained by the model.

4.2 Model Results

Table 1: Model Performance Comparison

Model	MSE	RMSE	R^2 Score
Linear Regression	1,395,098,000	37,351.02	0.6175
Random Forest	458,339,300	21,408.86	0.8743

Random Forest performed much better than Linear Regression. The Linear Regression model achieved an RMSE of 37,351.02 and an R^2 score of 0.6175. The Random Forest model achieved an RMSE of 21,408.86 and an R^2 score of 0.8743.

This means the Random Forest model explained about 87.43% of salary variation, while Linear Regression explained about 61.75%. Random Forest reduced the RMSE by about 42.7% compared to Linear Regression. This shows that AI salary prediction depends on nonlinear relationships between job features.

4.3 Salary and Skill Results

Table 2: Salary by Experience Level

Experience Level	Number of Jobs	Mean Salary	Median Salary
Entry Level	3,718	\$63,133.38	\$60,373.50
Mid Level	3,781	\$87,955.47	\$84,641.00
Senior Level	3,741	\$122,187.66	\$116,907.00
Executive Level	3,760	\$187,723.65	\$177,512.00

Experience level was the clearest salary driver. Executive-level roles have almost three times the average salary of entry-level roles.

Table 3: Salary by Company Size and Remote Work Ratio

Category	Number of Jobs	Mean Salary
Small Company	5,007	\$102,146.93
Medium Company	4,995	\$113,600.24
Large Company	4,998	\$130,322.45
0% Remote	5,075	\$114,140.23
50% Remote	5,005	\$115,776.81
100% Remote	4,920	\$116,160.55

Large companies had the highest average salary, followed by medium and small companies. Remote work had a smaller salary effect than experience level.

Table 4: Top Required Skills in AI Job Postings

Skill	Frequency
Python	4,450
SQL	3,407
TensorFlow	3,022
Kubernetes	3,009
Scala	2,794
PyTorch	2,777
Linux	2,705
Git	2,631
Java	2,578
GCP	2,442

Python was the most common skill, followed by SQL. Deep learning tools such as TensorFlow and PyTorch were also highly ranked. Kubernetes, Linux, Git, Java, Scala, and GCP show that AI roles often require software engineering and deployment skills, not only model-building skills.

4.4 Clustering Results

Table 5: K-Means Cluster Summary

Cluster	Mean Salary	Years Experience	Remote Ratio	Benefits Score
Cluster 0	\$88,308.34	3.49	84.03	7.96
Cluster 1	\$189,595.89	13.83	49.78	7.48
Cluster 2	\$86,951.53	3.36	15.95	7.09

Cluster 0 represents lower-experience jobs with high remote flexibility. Cluster 1 represents high-experience, high-salary jobs. Cluster 2 represents lower-experience jobs with low remote flexibility. The biggest difference between clusters is experience and salary.

5 Figures

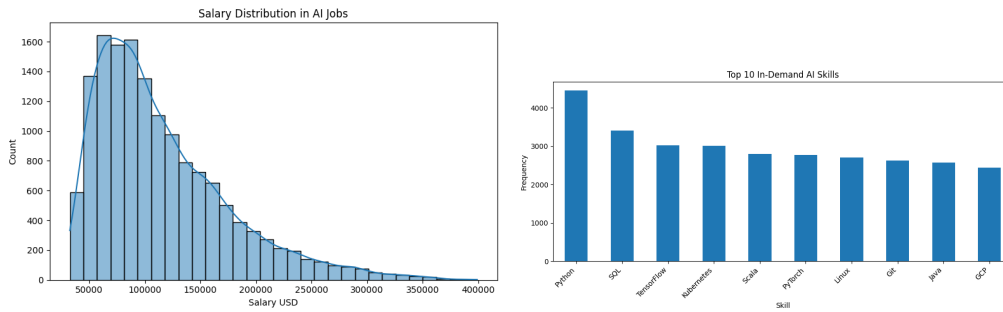


Figure 1: Salary distribution and top required AI skills. Most salaries fall in the middle range, while Python, SQL, TensorFlow, Kubernetes, and PyTorch are among the most common skills.

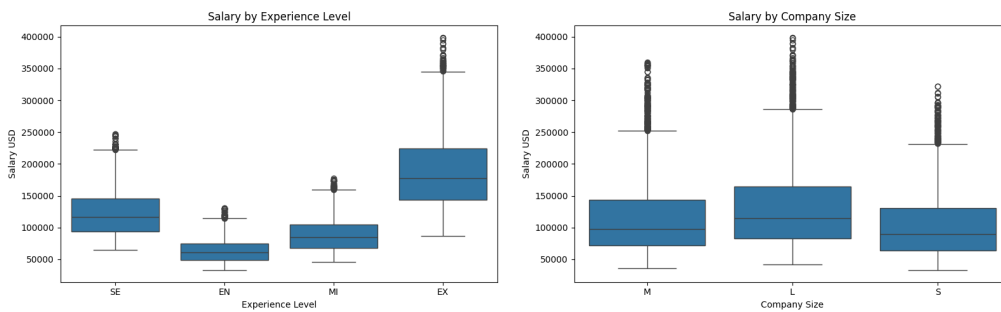


Figure 2: Salary patterns by experience level and company size. Salary increases strongly with experience level, and large companies have the highest average salaries.



Figure 3: Model prediction and clustering results. Random Forest predictions follow the actual salary trend, and K-Means separates jobs into meaningful salary and experience groups.

6 Discussion

The results show that experience level is the strongest factor influencing salary in the AI job market. Entry-level roles have the lowest average salary, while executive-level roles have the highest average

salary. This pattern is expected because senior and executive jobs usually require more technical skill, leadership ability, and industry experience.

Random Forest performed much better than Linear Regression, which shows that salary is not explained by one simple linear relationship. Instead, salary depends on several interacting factors such as experience level, job title, company size, industry, and required skills.

The skill analysis shows that AI jobs require a mix of programming, data, machine learning, cloud, and engineering skills. Python and SQL are the most common skills, while TensorFlow and PyTorch show the importance of deep learning frameworks. Kubernetes, Git, Linux, and GCP show that deployment and cloud skills are also valuable.

Company size also affects salary, with large companies offering the highest average pay. Remote work has only a small effect on salary, which suggests that remote work is more related to flexibility than compensation.

The clustering results show three useful AI job market segments: lower-experience remote-heavy roles, high-experience high-salary roles, and lower-experience onsite-heavy roles. This shows that clustering can reveal job market structure that is not obvious from regression alone.

7 Limitations

This project has several limitations. First, the dataset comes from Kaggle and may not fully represent the entire global AI job market. Some regions, companies, or industries may be overrepresented or underrepresented.

Second, salary depends on factors not fully included in the dataset, such as bonuses, stock options, negotiation, exact city, cost of living, company reputation, and team responsibilities.

Third, the skills column may contain different names for similar skills. For example, cloud skills may appear as AWS, Azure, GCP, or cloud computing, which can affect skill frequency counts.

Fourth, the project uses structured features but does not deeply analyze full job description text. A future version could use NLP methods such as TF-IDF or transformer embeddings.

Finally, K-Means clustering depends on the selected number of clusters. The three-cluster solution is useful, but a different value of k could produce different groups.

8 Conclusion

This project analyzed skill demand and salary trends in the Artificial Intelligence job market using 15,000 AI job postings. The analysis included data cleaning, visualization, regression modeling, feature importance, and clustering.

The main finding is that experience level is the strongest factor affecting salary. Entry-level jobs average about \$63,133, while executive-level jobs average about \$187,724. Company size also matters, with large companies offering higher average salaries than small and medium companies. Remote work has a smaller effect on salary.

The most demanded skills are Python, SQL, TensorFlow, Kubernetes, Scala, PyTorch, Linux, Git, Java, and GCP. This shows that AI professionals need programming, data, machine learning, cloud, and software engineering skills.

For salary prediction, Random Forest performed best with an RMSE of 21,408.86 and an R^2 score of 0.8743. This significantly outperformed Linear Regression. K-Means clustering also identified three meaningful job groups based on salary, experience, remote work, and benefits.

Overall, this project shows that data science can be used to better understand real AI career trends and help students prepare for the job market.

References

- [1] Sajjad, B. Global AI Job Market and Salary Trends 2025. Kaggle. <https://www.kaggle.com/datasets/bismasajjad/global-ai-job-market-and-salary-trends-2025>

- [2] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, E. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830, 2011.
- [3] McKinney, W. Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*, 2010.
- [4] Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., and Oliphant, T. E. Array programming with NumPy. *Nature*, 585, 357–362, 2020.
- [5] Hunter, J. D. Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95, 2007.
- [6] Waskom, M. L. Seaborn: Statistical Data Visualization. *Journal of Open Source Software*, 6(60), 3021, 2021.
- [7] MacQueen, J. Some Methods for Classification and Analysis of Multivariate Observations. *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1967.
- [8] Breiman, L. Random Forests. *Machine Learning*, 45, 5–32, 2001.
- [9] NeurIPS 2024 LaTeX Template. <https://www.overleaf.com/latex/templates/neurips-2024/tpsbbbrdqcmsh>